PROJECT DESIGN PHASE-II SPECIFICATION

StaffSync HRMS — Data Flow Diagrams & User Stories

Comprehensive architectural data movement specifications and agile user stories governing the cloud-native workforce management platform.

Project Name	**StaffSync — Enterprise Human Resource Management System**
System Architecture	3-Tier Decoupled Cloud (React 19 + Express 4 + MongoDB Atlas Cloud)
Version & Status	v1.0.0 (Production Release / Verified)
Project Lead / Author	Parikshit Sharma
Live Deployments	Client: staffsync-smoky.vercel.app | API: staffsync-czzq.onrender.com

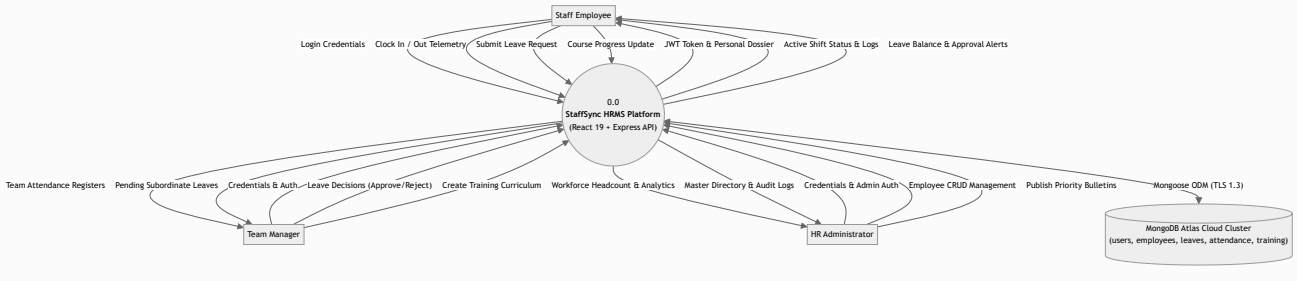
1. Overview & Data Flow Principles

A Data Flow Diagram (DFD) graphically depicts the flow of business information through the StaffSync platform. It traces data from external human actors (Staff Employees, Team Managers, HR Administrators) through stateless API boundary processes, validation and transformation algorithms, and into encrypted MongoDB Atlas persistence stores.

2. DFD Level 0: Context Level Diagram

The Level 0 Context Diagram establishes the global system boundary of the StaffSync platform and captures external inbound/outbound telemetry:

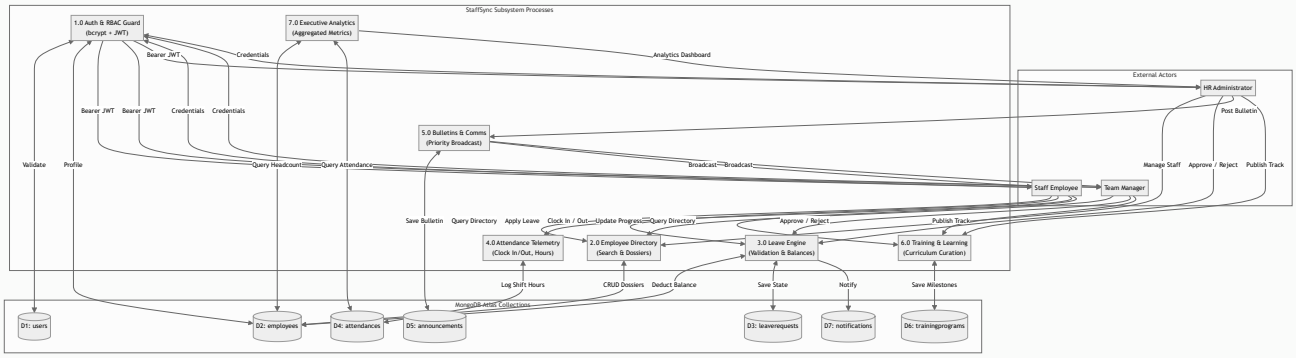
FIGURE 1: CONTEXT LEVEL DATA FLOW DIAGRAM (DFD LEVEL 0)



3. DFD Level 1: Subsystem Data Flow Diagram

The Level 1 diagram decomposes StaffSync into its 7 core transactional processes and explicit MongoDB data stores:

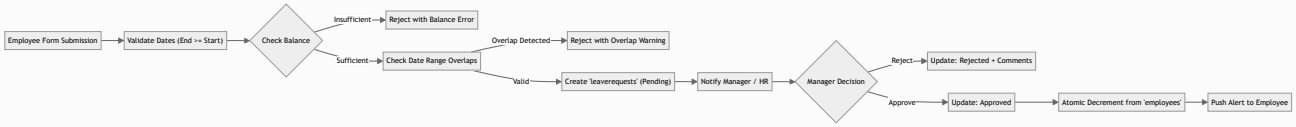
FIGURE 2: SUBSYSTEM DECOMPOSITION DATA FLOW DIAGRAM (DFD LEVEL 1)



4. DFD Level 2: Leave Lifecycle & Balance Accounting Flow

The Level 2 DFD specifies data integrity rules during leave submissions and manager approvals:

FIGURE 3: LEAVE REQUEST & BALANCE VERIFICATION PROCESS (DFD LEVEL 2)



5. User Stories & Acceptance Criteria Matrix

Agile user stories defining operational features, testable acceptance criteria, priorities, and sprint deliveries:

User Type	Epic / Category	User Story / Task	Acceptance Criteria	Priority	Release
All Users	User Authentication	As a user, I can log in using my registered work email and password.	Returns signed JWT token; redirects to matching role dashboard (/hr, /manager, /employee).	High	`v1.0.0`
All Users	User Registration	As a new staff member, I can register an account with initial department and role.	Creates record in users collection and provisions matching employee dossier.	High	`v1.0.0`
Staff Employee	Attendance Telemetry	As an employee, I can clock in and clock out with a single click.	Registers timestamp, activates live session stopwatch, and computes total shift hours.	High	`v1.0.0`
Staff Employee	Leave Application	As an employee, I can apply for leave and view real-time balance allowances.	Validates allowance before submitting; creates record with Pending status.	High	`v1.0.0`
Team Manager	Leave Approvals	As a manager, I can review, approve, or reject subordinate leave requests.	Approved leaves automatically decrement days from employee leave balances in real time.	High	`v1.0.0`
Team Manager	Skill Development	As a manager, I can curate and publish technical training tracks for my team.	Submits payload to POST /api/training; new track appears in team catalog.	High	`v1.0.0`
HR Admin	Workforce Directory	As an HR Admin, I can add, edit, and delete employee dossiers.	Changes persist directly to MongoDB Atlas; directory reloads dynamically without mock data.	High	`v1.0.0`
HR Admin	Bulletins & Comms	As an HR Admin, I can publish prioritized bulletins (High, Medium, General).	Bulletins display with priority color badges and are delivered to targeted audiences.	Medium	`v1.0.0`
All Users	Directory Search	As a user, I can search and filter colleagues by department and status.	Search bar instantly filters personnel cards without full page reloads.	Medium	`v1.0.0`
HR Admin	Workforce Analytics	As an executive, I can monitor org headcount, attendance, and leave volumes.	Metric cards render real-time calculations from live database collections.	Medium	`v1.0.0`

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